

Functional Regression-Based Multiple Imputation for Mixed Missing Data

Kexin Fu¹, Adam Ciarleglio¹

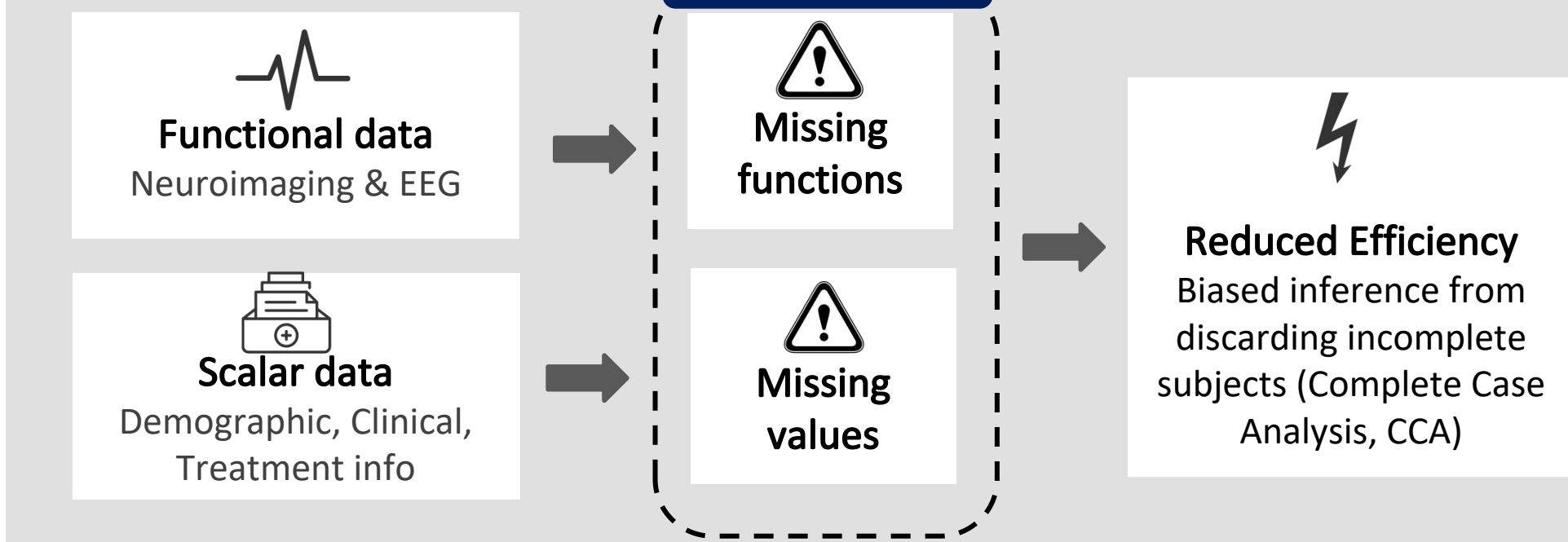
¹ Presenting author, ¹ Department of Biostatistics and Bioinformatics, Milken Institute School of Public Health

Context and Motivation

Biomedical Research

Modern biomedical research increasingly collects high-cost multimodal data, such as MRI, fMRI, and EEG, alongside clinical and demographic measures to better understand complex disease processes and health outcomes. These mixed data sources are often high-dimensional, noisy, and strongly correlated, which creates important analytic challenges.

Missing Data Challenge:



Regression-based Multiple Imputation (MI) is a promising strategy to handle Mixed Missingness

- Utilizing rich and Mixed Auxiliary information in biomedical studies
- Accounting for Uncertainty in imputed data
- An extension to Principle components-based regression maybe attractive improve stability and efficiency in the noisy biomedical settings.

fregMICE

We extended the *functional regression Multiple Imputation by Chained Equations (fregMICE)* framework by adding an FPCR-based approach for imputing missing scalar variables, alongside the existing PFR- approach for functional imputation, to handle mixed missing data. We aim to address the following research questions:

Primary

① Can updated fregMICE recover valid and efficient inference for coefficient functions under mixed missingness, compared to CCA?

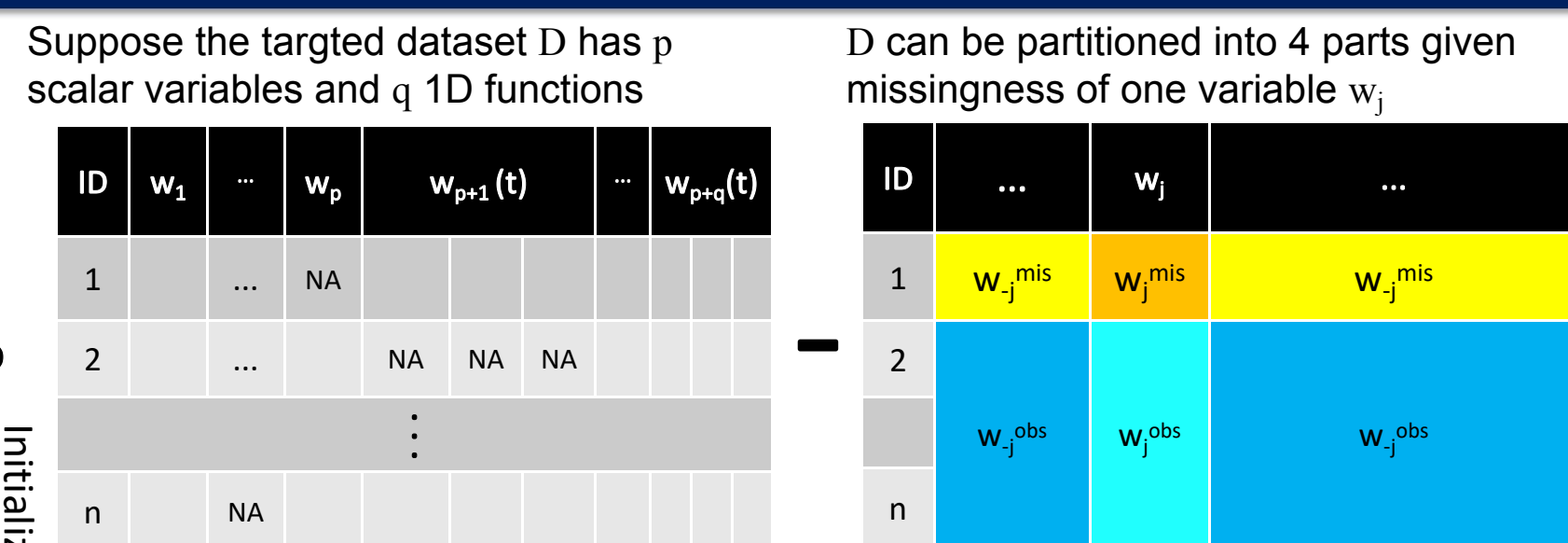
Secondary

- ② How do FPCR- and PFR-based imputation methods compare in recovering missing scalar values?
- ③ How sensitive are inferences to the choice of variance-covariance matrix in the model fit?

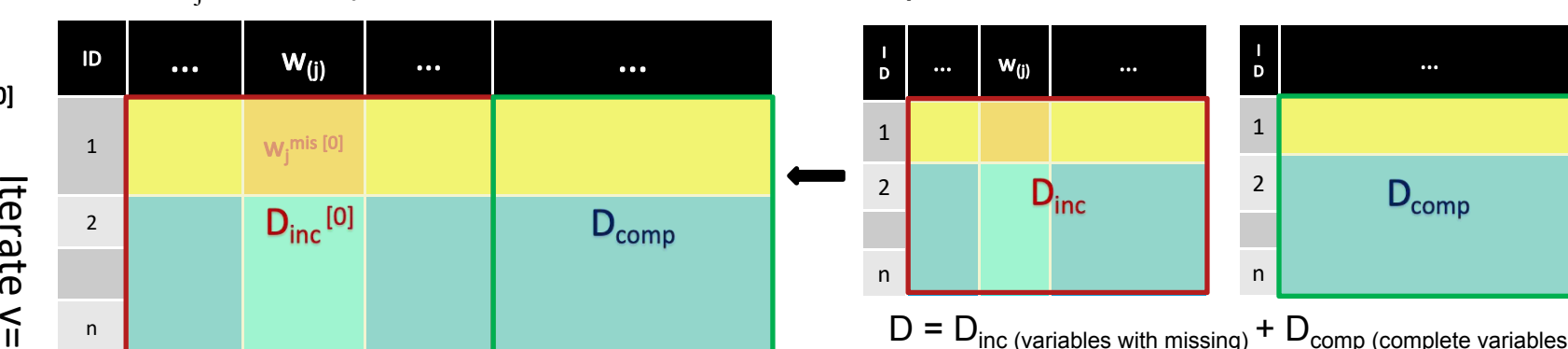
Methodology

fregMICE Imputation Pipeline

Data processing



0. For each variable j ($j=1 \dots p+q$), we initialize missing $w_j^{mis}[0]$ by randomly sampling from w_j^{obs} in D_{inc} . $D^{[0]}$ is the initial dataset for imputation.



1. To impute j th incomplete variable in v th iteration, we define the response vector (if j is scalar)/matrix (if j is function) and covariates matrix for imputation.

- $D_j^{[v]} = \{w_{j-1}^{[v]}, w_{j-1}^{[v]}, w_{j-1}^{[v-1]}, \dots, w_{p+q}^{[v-1]}\}$
- response vector/matrix, starting from the 1st iteration $w_j^{mis}[v-1] = w_j^{obs}[0]$
- covariates matrix, $w_{-j}^{obs}[v]$ are bootstrap samples from n_j^{obs}

In v th iteration when $j < J$, continue with $j=j+1$ imputation with $D_{j+1}^{[v]}$

Across $m=1 \dots M$:

scalar: $\hat{\theta} = \frac{1}{m} \sum_{m=1}^M \hat{\theta}^{(m)}$

function: $\hat{\beta}(t) = \frac{1}{m} \sum_{m=1}^M \hat{\beta}^{(m)}(t)$

Rubin's rule

scalar: $T_{\theta} = \bar{U}_{\theta} + (1 + \frac{1}{M})B_{\theta}$

function: $T_{\beta}(t) = \bar{U}_{\beta}(t) + (1 + \frac{1}{M})B_{\beta}(t)$

① Performance measure

mean squared error (MSE)/mean integrated squared error (MISE)

- mean pointwise (PW) SE

- 95% PW coverage

- 95% CI width

*2 What are functional regression models FPCR, PFR and PFFR?

Functional regression models characterize associations among scalar and functional variables; FPCR uses functional principal component scores as predictors, PFR uses penalized coefficient functions in scalar-on-function regression, and PFFR extends this framework to functional responses through flexible smooth effects.

$$\text{FPCR (1.1): } Y_i = \theta_0 + z_i^T \theta + \sum_{k=1}^q \sum_{a=1}^{A_k} \xi_{i,ka} \gamma_{ka} + \varepsilon_i \quad \text{FPCR (1.2): } Y_i \sim EF(\mu_i, \eta), \quad g(\mu_i) = \theta_0 + z_i^T \theta + \sum_{k=1}^q \sum_{a=1}^{A_k} \xi_{i,ka} \gamma_{ka}$$

$$\text{PFR (2.1): } Y_i = \theta_0 + z_i^T \theta + \sum_{k=1}^q \int X_{i,k}(t) \beta_k(t) dt + \varepsilon_i \quad \text{PFR (2.2): } Y_i \sim EF(\mu_i, \eta), \quad g(\mu_i) = \theta_0 + z_i^T \theta + \sum_{k=1}^q \int X_{i,k}(t) \beta_k(t) dt$$

$$\text{PFFR (3): } Y_i(t) \sim EF(\mu_i(t), \eta), \quad g\{\mu_i(t)\} = \beta_0(t) + \sum_{j=1}^p z_{i,j} \beta_j(t) + \sum_{k=1}^q \int X_{i,k}(s) \rho_k(s, t) ds$$

Y_i : scalar response
 $Y_i(t)$: functional response
 $z_i, z_{i,j}$: scalar predictor
 $X_{i,k}(t)$: functional predictor
 θ, γ_{ka} : scalar coefficient
 $\beta_k(t), \rho_k(s, t)$: coefficient functions

Simulation study

Data Generation

$$Y_i(t) = \beta_0(t) + \beta_1(t)z_{1i} + \beta_2(t)z_{2i} + \beta_3(t)z_{3i} + \varepsilon_i(t)$$

$$\text{Predictor: } z_{1i} \sim \text{Bin}(0.4) \quad (z_{2i}, z_{3i})^T \sim N\left(\begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.6 \\ 0 & 1 \end{pmatrix}\right)$$

$$\text{Coefficient: } \beta_2(t) = \frac{2}{\sqrt{2\pi}} e^{-t^2/2}, \beta_3(t) = \frac{-1}{\sqrt{2\pi}} \{e^{-(t-2)^2/2} + e^{-(t-9)^2/2}\}$$

$$\beta_0(t) = 0.25t, \beta_1(t) = \sin(\pi t/5)$$

Setting	Type	Key features
1	Bumpier	High-frequency sine waves & Gaussian peaks
2	Smoother	Low-frequency sine waves & exponential growth

Sample size: $n=100$ or 350 , $i=1 \dots n$

Missing proportion: 10%, 20%, 30%

Missing scenario

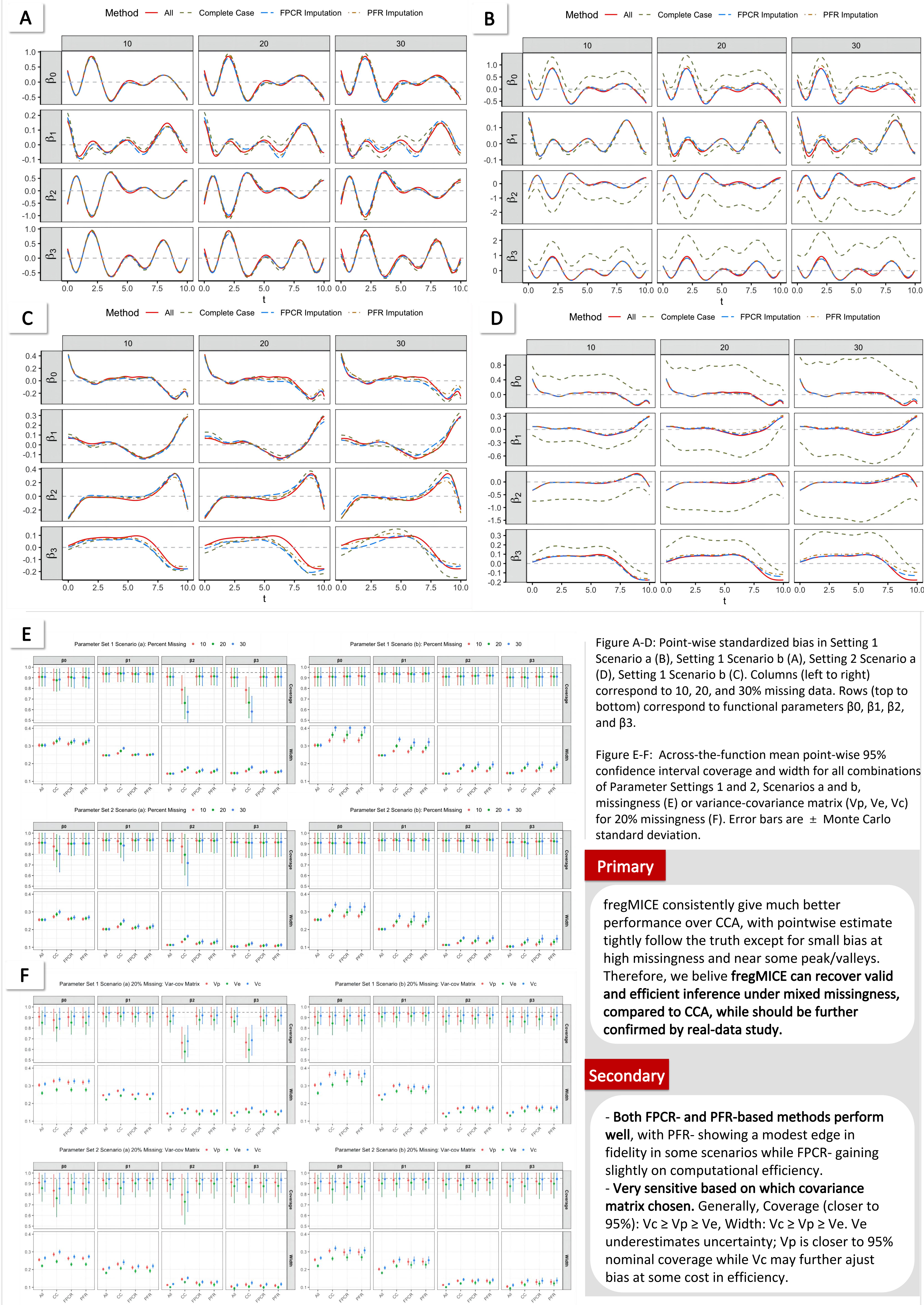
(a): scalar missing $\logit(P(z_{2i} = 1)) = 10 - 20I\{s_i > a_1\}$

(b): mixed missing

$$\logit(P(z_{2i} = 1)) = a_0 + \frac{1}{1+e^{-z_{1i}}} - \frac{1}{1+e^{-z_{3i}}} \logit(P(R_{Yi} = 1)) = \psi_0 + \frac{1}{1+e^{-z_{1i}}} - \frac{1}{1+e^{-z_{3i}}}$$

Evaluation:

- 500 replicates with reproducible seeds using GW HPC (Pegasus).
- fregMICE (PFR vs. FPCR), Complete-Case, and ANM (Benchmark).
- Analytic model applied PFFR across all for functional response.



Primary

fregMICE consistently give much better performance over CCA, with pointwise estimate tightly follow the truth except for small bias at high missingness and near some peak/valleys. Therefore, we believe fregMICE can recover valid and efficient inference under mixed missingness, compared to CCA, while should be further confirmed by real-data study.

Secondary

- Both FPCR- and PFR-based methods perform well, with PFR- showing a modest edge in fidelity in some scenarios while FPCR- gaining slightly on computational efficiency.
- Very sensitive based on which covariance matrix chosen. Generally, Coverage (closer to 95%): $V_c \geq V_p \geq V_e$, Width: $V_c \geq V_p \geq V_e$. Ve underestimates uncertainty; Vp is closer to 95% nominal coverage while Vc may further adjust bias at some cost in efficiency.